

The value (s) of k for which the matrix

$$\begin{bmatrix} -1 & 1 & k \\ 0 & 1 & 4 \\ k & 3 & 7 \end{bmatrix}$$

is nonsingular is/are given by:

- ☐ $k \neq -1$ or $k \neq 0$
☐ $k \neq -1$
☐ $k \neq 5$ or $k \neq 1$
☐ $k \neq 5$ and $k \neq 1$
☐ $k \neq 5$
☒ None of the given option
☐ $k \neq -1$ or $k \neq 2$

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Suppose that we are given two square matrices A and B of the same size. Then find an expression for $(A + B)^2$ whenever A and B commute.

- ☐ $A^2 + B^2$
☐ $A^2 + B^2 - 2AB$
☐ None of the given options
☒ $A^2 + B^2 + 2AB$
☐ $A^2 + B^2 + AB$

[Clear my choice](#)

Question 2
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Consider the matrices below with $X = AB$, where

$$A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 3 & -1 & 1 & 3 \\ -2 & 5 & -k & k+1 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 4 & 1 \\ 3 & -1 & 1 \\ -2 & 5 & k(k-1) \end{bmatrix}$$

For which value(s) of k will the entry X_{23} be zero?

- ☐ None of the given option
☐ $k = 0$ or $k = 1$
☒ $k = \frac{1 \pm \sqrt{5}}{2}$
☐ No such k
☐ $k = -2$
☐ $k = 3$

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Question 3
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Question 4

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Consider the system of linear equations with unknown a , b , and c and λ is parameter to be determined.

$$\begin{cases} a + (\lambda^2 + 1)b - (\lambda^2 - \lambda - 8)c = 3 - \lambda, \\ a - (\lambda^2 + 1)b = 3 \\ -(\lambda^2 + 1)b - (7 + \lambda - \lambda^2)c = \lambda. \end{cases}$$

Use elementary row operation to solve the above system. Find the value(s) for λ for which the system has no solution?

- ☐ None of the given options
- ☐ $\lambda = 2$ or $\lambda = -3$
- ☒ $\lambda = -2$ or $\lambda = 3$
- ☐ $\lambda = -2$ and $\lambda = -3$

[Clear my choice](#)

Question 5

Not yet answered

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Consider the matrices

$$X = \begin{bmatrix} x + y - 1 & 0 \\ 0 & 3 \end{bmatrix} \text{ and } Y = \begin{bmatrix} 5 & 0 \\ 0 & 2x - 3y - 7 \end{bmatrix}$$

The value(s) of x for which the matrices X has no inverse

and $Y = -5I_2$ is/are given by:

- ☐ $x = 0$
- ☐ $x = -1$
- ☐ No such value of x exists
- ☒ $x = 1$

[Clear my choice](#)

Question 6

Not yet answered

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Let C be an $m \times n$ matrix and k a scalar, prove that if $kC = O$ then either $k = 0$ or $C = O$ where O is the zero matrix with zero entries everywhere.

- ☒ Let $kC = O$, where $C = [C_{ij}]$ with $1 \leq i \leq m$ and $1 \leq j \leq n$. Then by identification, we have $kC_{ij} = 0$ for $1 \leq i \leq m$ and $1 \leq j \leq n$
- ☒ This implies that $k = 0$ or $C = O$ for $1 \leq i \leq m$ and $1 \leq j \leq n$
- ☐ If there exist i or j such that $C_{ij} \neq 0$ Then $k = 0$ which implies that $kC = O$.
- ☐ Otherwise, $C_{ij} = 0$ for $1 \leq i \leq m$ and $1 \leq j \leq n$ which implies that $C = O$ is indeed the zero matrix. This conclude the desired proof.
- ☐ None of the given options is correct

Question 7

Not yet answered

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Let A be the coefficient matrix of the system

$$3x + 4y + z = 8$$

$$-2x + 7y + z = 1$$

$$8x + y + 5z = -2$$

Now add the following cofactor entries (of A) C_{11} , C_{32} and $-C_{33}$ to get the resulting sum:

- ☐ None of the given option
- ☐ -1
- ☐ -5
- ☒ 0

[Clear my choice](#)

Let A and B be square matrix of the same size. Which of the following statement is correct?

- ☒ AA^T is skew-symmetric
- ☐ $(A + A^T)^T$ is symmetric
- ☐ $A + A^T$ is skew-symmetric
- ☐ None of the given options
- ☐ $(A - B)^T$ is symmetric

[Clear my choice](#)

Question 8

Not yet answered

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Label whether the given matrix is row echelon form or reduced row echelon form.

$$\begin{bmatrix} 1 & -3 & 0 & 3 & 0 \\ 0 & 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

- ☐ None of the given option
- ☒ This matrix is in reduced row echelon form or This matrix is in row echelon form
- ☒ This matrix is in reduced row echelon form
- ☐ This matrix is not in row echelon form

Question 9

Not yet answered

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Use inspection to find the determinant of $C = AB$, where

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -3 \end{bmatrix}$$

- ☒ 3
- ☐ -1
- ☐ -3
- ☐ 2
- ☐ None of the given options

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Assume that A is a 2×2 matrix with integer entries. If B is the inverse of A and B has integer entries, then choose the correct statement:

- ☐ $\det(A) = 1$ or $\det(A) = 0$ and B has integer entries
- ☒ None of the given options
- ☐ $\det(A) = -1$ or $\det(A) = 0$
- ☐ $\det(A) = -1$ or $\det(A) = 1$
- ☐ $\det(A) = -1$ and $\det(A) = 0$

[Clear my choice](#)

Question 11

Not yet answered

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Question 12

Not yet answered

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Suppose L is the line with symmetric equations

$$x - 3 = y - 1 = z + 2.$$

Now determine whether the above line is parallel to the plane

$$x + y - 2z = 0.$$

Choose the correct answer:

- ☐ The direction vector for L is perpendicular to the normal vector $\vec{n} = \langle 1, 1, -2 \rangle$
- ☐ None of the given option
- ☐ The line L is perpendicular to the given plane with normal vector $\vec{n} = \langle 1, 1, -2 \rangle$
- ☐ The line L is parallel to the normal vector of the given plan
- ☒ The dot product of direction vector for the line L and the normal vector for the plane is not equal to zero

[Clear my choice](#)

Question 13

Not yet answered

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The augmented matrices of the three systems are given below.

$$\left[\begin{array}{ccc|c} 1 & a & 0 & 4 \\ 1 & 0 & 1 & -2 \\ 0 & 0 & b & 1 \\ 0 & 0 & 0 & 0 \end{array} \right] \text{ where } a, b \in \mathbb{R} \text{ with } b \neq 0.$$

How many solutions are there in this case?

- ☒ Infinitely many solutions
- ☐ None of the given options is true
- ☐ All of the given options are correct
- ☐ No solutions
- ☐ Exactly one solution

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Question 14

Not yet answered

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We consider the two points $P(1, 2, 1)$ and $Q(2, -1, 1)$.

Determine the symmetric equations of the line passing through the points P and Q . Assume that $R(3, -4, -3)$ is another point. Assume you want to show using the symmetric equations of the line that the three points are collinear. Choose the correct statement.

- ☐ The three point P , Q , and R are not collinear at all.
- ☐ No such statement.
- ☐ The three point P , Q , and R are collinear whenever R does not belong to the line passing through P and Q .
- ☒ The three point P , Q , and R are collinear whenever R belong to the line passing through P and Q .

[Clear my choice](#)

Question 15

Not yet answered

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Find the symmetric equations of the line L that passes through the point $P(-1, 2, -3)$ and is parallel to the vector

$$\vec{U} = \langle 1, -2, -1 \rangle.$$

Find the point in which the required line L intersects with the coordinate planes.

- ☐ The point of intersections of L with the xy -plane is $-4(1, 0, 0)$.
- ☒ The point of intersections of L with the xy -plane is $-4(0, -2, 0)$.
- ☐ None of the given options
- ☐ The point of intersections of L with the xy -plane is $-4(1, -2, 0)$.
- ☐ The point of intersections of L with the xy -plane is $4(1, -2, 0)$.

[Clear my choice](#)

Let $\vec{u} = \langle 0, 2, 2 \rangle$, $\vec{v} = \langle 2, 3, 0 \rangle$ and $\vec{w} = \langle 2, 0, 1 \rangle$ be three vectors in standard position.
Then, the equation of the plane parallel to vectors $\vec{v}$ and $\vec{w}$ and passing through the tip of $\vec{u}$ is given by:

- ☒ $3x - 2y - 6z = -16$
- ☐ $2x - 4y - 6z = -8$
- ☐ $x - 2y - 6z = 1$
- ☐ $3x - 2x - 6z = -2$
- ☐ None of the given options

[Clear my choice](#)

Question **16**
Not yet answered
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Find the area of the triangle determined by its three vertices $P(0, -1, 1)$, $Q(1, -2, 1)$, and $R(1, 0, 1)$. The area of the triangle PQR is given by:

- ☐ 8 units²
- ☐ 2 units²
- ☐ 7 units²
- ☒ None of the given options
- ☐ 5 units²

[Clear my choice](#)

Question **17**
Not yet answered
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Let α be a complex number. Use the Euler form of a complex number to solve for α from the expression
 $1^n = -1$

- ☐ $\alpha = \frac{k+1}{2n}$ for $k, n \in \mathbb{Z}$ with $n \neq 0$
- ☒ $\alpha = \frac{2k}{-n+1}$ for $k, n \in \mathbb{Z}$ with $n \neq 1$
- ☐ None of the given options is correct.
- ☐ $\alpha = \frac{2k+1}{2n}$ for $k, n \in \mathbb{Z}$ with $n \neq 0$
- ☐ $\alpha = \frac{k+1}{n+1}$ for $k, n \in \mathbb{Z}$ with $n \neq 0$

[Clear my choice](#)

Question **18**
Not yet answered
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Let
 $z_1 = 1 - 2i - i^2$
and
 $z_2 = x + y + ix$
be two complex numbers such that
 $z_1 = z_2$.
Solve for x and y from which you deduce that the product xy is equal to

- ☐ 8
- ☒ -8
- ☐ -2
- ☐ None of the given options
- ☐ 4

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Question **19**
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Question **20**Not yet answered
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Let L be the line that passes through the points $A(-1, 1, -1)$ and $B(1, -1, 2)$, and V the plane containing L and the point $C(1, 0, 1)$. The parallel vectors $\vec{u}$ and $\vec{v}$ lying in the plane V are given by:

- ☐ $\vec{u} = (0, 1, -1)$ and $\vec{v} = (2, -1, -2)$
☒ $\vec{u} = (0, 1, -1)$ and $\vec{v} = (2, -1, 2)$
☐ $\vec{u} = (1, 1, -1)$ and $\vec{v} = (2, -1, 2)$
☐ $\vec{u} = (2, -1, 3)$ and $\vec{v} = (0, 1, -1)$
☐ None of the given options

Clear my choice

Question **21**Not yet answered
Marked out of 4.00

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Let L be the line that passes through the points $A(-1, 1, -1)$ and $B(1, -1, 2)$, and V the plane containing L and the point $C(1, 0, 1)$. First, determine two parallel vectors $\vec{u}$ and $\vec{v}$ that lying in the plane V . Then, the Cartesian equation of the plane is given by:

- ☐ $x - 2y - 2z - 1 = 0$
☐ $x + 2y - 2z - 1 = 0$
☐ $x - 2y + 2z - 1 = 0$
☒ $x - 2y - 2z + 1 = 0$
☐ None of the given options

Clear my choice

Question **22**Not yet answered
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Let $P(z) = z^2 - 2z + 4$ and assume that the root of $P(z)$ are given by α and β . The expression for $\alpha^n + \beta^n$ is given by

- ☐ None of the given options
☒ 16 for $n = 3$
☐ -16 for $n = 3$
☐ $-8i$ for $n = 3$
☐ -8 for $n = 3$

Clear my choice

Question **23**Not yet answered
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What are the cube roots of $8i$?

- ☐ $z_0 = 2 \left[\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right], z_1 = 2 \left[\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right], z_2 = 2 \left[\cos \frac{\pi}{2} - i \sin \frac{\pi}{2} \right]$
☒ $z_0 = 2 \left[\cos \frac{\pi}{6} - i \sin \frac{\pi}{6} \right], z_1 = 2 \left[\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right], z_2 = 2 \left[\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right]$
☐ $z_0 = 2 \left[\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right], z_1 = 2 \left[\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right], z_2 = 2 \left[\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right]$
☐ $z_0 = 2 \left[\cos \frac{\pi}{6} - i \sin \frac{\pi}{6} \right], z_1 = 2 \left[\cos \frac{5\pi}{6} - i \sin \frac{5\pi}{6} \right], z_2 = 2 \left[\cos \frac{\pi}{2} - i \sin \frac{\pi}{2} \right]$
☐ None of the given options

Clear my choice

Let

$$z = 2 - 5i.$$

Then $|z|^2$ is given by

- ☐ 2
- ☒ 29
- ☐ None of the given options
- ☐ 30
- ☐ 6

[Clear my choice](#)

Question 24

Not yet answered

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Consider that z is a real number, determine the value (s) for β such that

$$z = \frac{\beta + 2i}{1 + i\beta}$$

- ☐ None of the given options
- ☐ $\beta = \pm 1$
- ☒ $\beta = \pm 2$
- ☐ $\beta = \pm 3$
- ☐ $\beta = \pm\sqrt{2}$

[Clear my choice](#)

Question 25

Not yet answered

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Consider the matrices

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 2 & 5 \\ 1 & 6 \end{bmatrix}, C = \begin{bmatrix} 2 & 5 \\ 3 & -1 \end{bmatrix}$$

Compute the matrices AB and BC and conclude:

- ☐ $AB = \begin{bmatrix} 2 & -5 \\ 0 & 0 \end{bmatrix}$ and $BC = \begin{bmatrix} 2 & 5 \\ 0 & 0 \end{bmatrix}$ but $B \neq C$.
- ☒ None of the given options
- ☐ $AB = \begin{bmatrix} 2 & 5 \\ 0 & 0 \end{bmatrix}$ and $BC = \begin{bmatrix} 2 & 5 \\ 0 & 0 \end{bmatrix}$ but $B \neq C$.
- ☐ $AB \neq BC$ but $B = C$.
- ☐ $AB = BC = \begin{bmatrix} 2 & 5 \\ 0 & 0 \end{bmatrix}$ but $B = -C$.

[Clear my choice](#)

Question 26

Not yet answered

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Let

$$(3A^T - I_2)^{-1} = \begin{bmatrix} 2 & 2 \\ 3 & 4 \end{bmatrix}$$

Then the matrix A is given by:

- ☒ $A = \begin{bmatrix} 1 & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} \end{bmatrix}$
- ☐ $A = \begin{bmatrix} 1 & -\frac{1}{3} \\ -\frac{1}{2} & \frac{2}{3} \end{bmatrix}$
- ☐ $A = \begin{bmatrix} \frac{1}{3} & -\frac{1}{2} \\ -\frac{1}{3} & \frac{1}{3} \end{bmatrix}$
- ☐ $A = \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{2} & -\frac{1}{3} \end{bmatrix}$
- ☐ None of the given options are correct.
- ☐ $A = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ -\frac{1}{2} & \frac{1}{3} \end{bmatrix}$
- ☐ $A = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ -\frac{1}{2} & -\frac{1}{3} \end{bmatrix}$

Question 27

Not yet answered

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